

- Consider the Classical solⁿ of a string of length L , the Energy E and angular Momentum $J \equiv |J^{12}|$ are given by

$$E = \frac{1}{4\alpha'} L$$

(Energy)

$$J = L^2 / 16\alpha'$$

(Angular Mom.)

→ Rotating string

$$L \Rightarrow \alpha' E^2 = J$$

* When we Quantize the theory we have

$$J = n\hbar \quad \text{and so}$$

$$\alpha' E^2 = n\hbar, \quad E = (\text{Rest mass})$$

⇒ We have a series of masses m_n

$$\alpha' m_n^2 = n\hbar$$

We have an ∞ # of particles whose m_n^2 is proportional to their spin J .

The Old Covariant Method:

$$\left\{ \begin{array}{l} \text{Poisson} \\ \text{Dirac} \end{array} \right. \quad \left\{ \begin{array}{l} \{A, B\} = C \\ [A, B] = i\hbar C \end{array} \right.$$

→ Can always be applied to phase space coordinates

(i) Momenta: $P^\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}_\mu}$

In the (conformally) flat gauge $g^{\alpha\beta} = \eta^{\alpha\beta}$

$$P^\mu = \frac{1}{2\alpha'} \dot{x}^\mu$$

• We find $\{\bar{x}^\mu(\sigma), \bar{x}^\nu(\sigma')\} = \alpha' \{P^\mu(\sigma), P^\nu(\sigma')\}$

$$\{\bar{x}^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

Which are all classical constraints; thus, we can impose the Quantum Dirac relations:

$$[\bar{x}^\mu(\sigma), \bar{x}^\nu(\sigma')] = 0 = [P^\mu(\sigma), P^\nu(\sigma')] = 0$$

$$[\tilde{X}^{\mu}(\sigma), P^{\nu}(\sigma')] = i\hbar \delta(\sigma - \sigma') \eta^{\mu\nu}$$

* The Schrödinger representation:

$$\begin{aligned} \tilde{X}^{\mu}(\sigma) &\mapsto \hat{\tilde{X}}^{\mu}(\sigma) \\ P^{\mu}(\sigma) &\mapsto -i\hbar \frac{\delta}{\delta X^{\mu}(\sigma)} \end{aligned}$$

where

$$\delta X^{\nu}(\sigma) / \delta X^{\mu}(\sigma') = \delta^{\nu}_{\mu} \delta(\sigma - \sigma')$$

Constraints:

(i) Open string: $T_{\alpha\beta} = 0$ or $L_n = 0$
why?

$$T(\tau) = \sum_{n \in \mathbb{Z}} L_n \tau^{n-2}$$

and since $T(\tau) = 0$ & $T(\bar{\tau}) = 0$

then $T(\tau) = 0 \Leftrightarrow L_n = 0$

(ii) Closed string: $L_n = 0$ and $\bar{L}_n = 0$

These are the constraints which we impose on a wave fun $\psi[X^{\mu}(\tau, \sigma)] \equiv \langle X^{\mu}(\sigma, \tau) | \psi \rangle$

The Open string:

* In extending $\sigma \rightarrow [-\pi, \pi]$ we combined both constraints into a single condition

$$L_n = 0 \quad \forall n$$

(Recall $X_{-n}^{\mu} = X_n^{\mu}$, and so " $\bar{L}$ " = L here).

- A constant must be added to L_0 to parametrize the arbitrariness of the ordering chosen. So, at best we may have $(L_0 - a)\psi = 0$

or, normalizing $a = 1$

$$(L_0 - 1)\psi = 0, \quad L_n \psi = 0, \quad n \geq 1$$

(Physical state conditions)

$$L_n^\dagger = L_{-n}, \text{ it's clear } \langle \psi | L_n | \psi \rangle = 0, \quad n \neq 0$$

$$\bullet \quad \Sigma^\mu, P^\mu \leftrightarrow q^\mu, \alpha_0^\mu = \sqrt{2\alpha'} p^\mu, \alpha_n^\mu, \quad n \neq 0$$

$$\bullet \quad \text{Use } \hbar = 1, \{ \alpha_n^\mu, \alpha_m^\nu \} = -i\eta \delta_{n+m,0} \eta^{\mu\nu}$$

$$\text{and so } \{ \cdot, \cdot \} \rightarrow [\cdot, \cdot] \propto -i$$

$$[\alpha_n^\mu, \alpha_m^\nu] = n \delta_{n+m,0} \eta^{\mu\nu}$$

$$36, 52$$

$$2$$

$$X_0^\mu(\sigma) = \frac{1}{\alpha'} \int_0^\pi d\sigma' X^\mu(\sigma, \sigma')$$

$$\{X^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

Integrating over σ : at $t=0$

$$\left\{ \frac{1}{\pi} g^{\mu\nu}, P^\nu \right\} = \eta^{\mu\nu}$$

$$\int_{-\pi}^{\pi} d\sigma' \{ \cdot, \cdot \} = \eta^{\mu\nu}$$

$$\{ q^\mu, P^\mu \} = \eta^{\mu\nu}$$

$$\Rightarrow \{ q^\mu, P^\mu \} = \eta^{\mu\nu}$$

$$\text{Thus: } [q^\mu, P^\mu] = i\eta^{\mu\nu}$$

This is embedded in the ladder ops

$$\alpha_0^\mu = \sqrt{2\alpha'} p^\mu = -i\sqrt{2\alpha'} \frac{\partial}{\partial q^\mu}$$

$$X_0^\mu = q^\mu$$

Since $P^\mu(\sigma)$ is self adjoint $P^{\mu\dagger} = P^\mu$,

$$\text{then } (\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$$

* We can rederive these relations as follows:

$$\begin{aligned} P^\mu(\sigma) &= \sqrt{2\alpha'} P^\mu + \frac{X^{\mu'}(\sigma)}{\sqrt{2\alpha'}} \\ &= -i\sqrt{2\alpha'} \frac{\partial X^\mu(\sigma)}{\partial X^\mu(\sigma)} + \frac{X^{\mu'}(\sigma)}{\sqrt{2\alpha'}} = \sum_{n=-\infty}^{\infty} \alpha_n^\mu e^{in\sigma} \end{aligned}$$

$$\frac{\delta}{\delta X^\mu(\sigma)} = \sum_{n=0}^{\infty} \frac{\partial X_n^\mu}{\partial X^\mu(\sigma)} \frac{\partial}{\partial X_n^\mu}$$

$$\begin{aligned}
 X_0^\nu(\tau) &= \frac{i}{\pi} \int_0^{2\pi} d\sigma' \mathcal{X}(\tau, \sigma') \\
 &= \sum_{n=-\infty}^{\infty} \frac{2}{\pi} \int_0^{2\pi} d\sigma' \cos(n\sigma') \delta_n^\nu \delta(\sigma - \sigma') \\
 &= \sum_{n=-\infty}^{\infty} \frac{2}{\pi} \cos(n\sigma) \frac{\partial}{\partial x_n^\mu} \\
 &= \frac{2}{\pi} \sum_{n \neq 0} (e^{in\sigma} + e^{-in\sigma}) \frac{\partial}{\partial x_n^\mu}
 \end{aligned}$$

Note, $x_{-n}^\mu = x_n^\mu$, thus

$$\frac{2}{\pi} \left(\sum_{n=1}^{\infty} e^{in\sigma} \frac{\partial}{\partial x_n^\mu} + \sum_{n=1}^{\infty} e^{-in\sigma} \frac{\partial}{\partial x_{-n}^\mu} \right)$$

$$\therefore \frac{\delta}{\delta x_n^\mu(\sigma)} = \frac{2}{\pi} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\partial}{\partial x_n^\mu}$$

Subbing this into $\mathcal{P}^\mu = -i\sqrt{2\alpha'} \pi \frac{\delta}{\delta \dot{x}_\mu(\sigma)} + \frac{\dot{x}^\mu}{\sqrt{2\alpha'}}$

$$-i\sqrt{2\alpha'} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\partial}{\partial x_n^\mu} + \dot{x}^\mu(\sigma)/\sqrt{2\alpha'}$$

For $\dot{x}^\mu$ we'd like to express this in terms of modes x_n^μ :

$$\begin{aligned}
 \mathcal{X}^\mu(\tau, \sigma) &= \sum_{n=-\infty}^{\infty} x_n^\mu(\tau) \cos(n\sigma) \\
 &= x_0^\mu(\tau) + \frac{i}{2} \sum_{n=1}^{\infty} x_n^\mu(\tau) e^{in\sigma}
 \end{aligned}$$

$$\therefore \mathcal{X}^\mu(\tau, \sigma) = \frac{i}{2} \sum_{n \neq 0} x_n^\mu(\tau) n e^{in\sigma}$$

Note, we can include $n=0$ since $n=0$ really is $=0$

$$\mathcal{X}^\mu(\tau, \sigma) = \frac{i}{2} \sum_{n=-\infty}^{\infty} x_n^\mu(\tau) e^{in\sigma}$$

$$\begin{aligned}
 \mathcal{P}^\mu(\sigma) &= -i\sqrt{2\alpha'} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\partial}{\partial x_n^\mu} + \frac{i}{2} \sum_{n=-\infty}^{\infty} x_n^\mu n e^{in\sigma} \\
 &= \sum_n \alpha_n^\mu e^{in\sigma}
 \end{aligned}$$

$$\alpha_n^\mu = -i \left(\sqrt{2\alpha'} \frac{\partial}{\partial x_{-n}^\mu} + \frac{x_n^\mu n}{2\sqrt{2\alpha'}} \right)$$

* Virasoro ops: $L_n = \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha_m^\mu \alpha_{n-m}^\mu$

Normal ordering:

$$:\alpha_n^\mu \alpha_m^\nu: \equiv \begin{cases} \alpha_n^\mu \alpha_m^\nu, & \text{if } m > n \\ \alpha_m^\nu \alpha_n^\mu, & \text{if } n > m \end{cases}$$

i.e. we place the "larger" operator to the right

- Take a vacuum state $|0, p\rangle$:

$$\alpha_n^\mu |0, p\rangle = 0, \quad n \geq 1$$

$$\alpha_0^\mu |0, p\rangle = \sqrt{2\alpha'} p^\mu |0, p\rangle$$

Thus, for the present purpose only α_n^μ needs to be Normal ordered. Because of the normal ordering issue the Witt Alg. becomes the full:

$$[L_n, L_m] = (n-m) L_{n+m} + \frac{D}{24} n(n^2-1) \delta_{n+m,0}$$

Take $m \rightarrow -n, \quad n \geq 0$

$$[L_n, L_{-n}] = 2n L_0 + \frac{D}{12} n(n^2-1) \mathbb{1}$$

$$\langle 0,0 | [L_n, L_{-n}] | 0,0 \rangle = \langle 0,0 | L_0 | 0,0 \rangle + \frac{D}{12} n(n^2-1) \langle 0,0 | 0,0 \rangle$$

$$\langle 0 | [L_n, L_{-n}] | 0 \rangle = \langle 0,0 | L_n L_{-n} | 0,0 \rangle$$

$$L_n = \frac{1}{2} : \sum_m \alpha_m^\mu \alpha_{n-m}^\nu \eta_{\mu\nu} :$$

$$\langle 0,0 | \frac{1}{2} : \sum_m \alpha_m^\mu \alpha_{n-m}^\nu \eta_{\mu\nu} : \frac{1}{2} : \sum_l \alpha_l^\sigma \alpha_{-n-l}^\lambda \eta_{\sigma\lambda} : | 0,0 \rangle$$

All states aside from the zero modes remain:

$$\frac{1}{4} \sum_{m=1}^{n-1} m(n-m) \langle 0,0 | 0,0 \rangle$$

$$\begin{aligned} L_0 &= \frac{1}{2} \sum_m \alpha_m^\mu \alpha_{-m}^\nu \eta_{\mu\nu} = \frac{1}{2} \left(\sum_{m=1}^{\infty} \alpha_m^\mu \alpha_{-m}^\mu + \sum_{m=1}^{\infty} \alpha_m^\mu \alpha_{-m}^\mu \right) \\ &= \frac{1}{2} \alpha_0^\mu \alpha_0^\mu \eta_{\mu\nu} + \frac{1}{2} \sum_{n=1}^{\infty} : \alpha_{-n}^\mu \alpha_n^\mu : + \sum_{n=1}^{\infty} : \alpha_n^\mu \alpha_{-n}^\mu : \\ &= \frac{1}{2} \alpha_0 \cdot \alpha_0 + \sum_{n=1}^{\infty} (\alpha_{-n} \cdot \alpha_n) \end{aligned}$$

$$\langle x^\mu(0, \epsilon) | L_0 | \psi \rangle = \langle x^\mu(0, 0) | L_0 | \psi \rangle \langle 0, 0 | L_0 | 0, 0 \rangle$$

$$\langle 0, 0 | \psi \rangle$$

$$\Rightarrow L_0 \psi = 0$$

$$(2\alpha L_0 + \frac{2}{12} D \alpha (n^2 - 1)) \psi = 0$$

$$n=0, (\text{trivial})$$

$$n=1: L_0 \psi = 0 \quad (\text{trivial})$$

$$n=-1: L_0 \psi = 0 \quad (\text{trivial})$$

$$n \geq 1: 2L_0 \psi = -\frac{D}{12} (n^2 - 1) \psi$$

$$\text{or } L_0 \psi + \frac{D}{24} (n^2 - 1) \psi = 0, \quad n \neq \pm 1$$

* Recall,

$$(\alpha_n^\mu)^\dagger = (\alpha_{-n}^\mu)$$

thus,

$$L_0 = \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + \sum_{n=1}^{\infty} \alpha_n^\mu \alpha_{-n\mu} \equiv \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + N$$

$$N \equiv \sum_{n=1}^{\infty} (\alpha_n^\mu)^\dagger \alpha_{n\mu}$$

$$L_1 = L_{-1} = \alpha_0^\mu \alpha_{1\mu} + \sum_{n=1}^{\infty} (\alpha_n^\mu)^\dagger \alpha_{n+1,\mu}$$

$$\alpha_0^\mu = -i\sqrt{2\alpha'} \frac{\partial}{\partial x_0^\mu} \equiv \sqrt{2\alpha'} p^\mu$$

$\rightarrow \alpha_0^\mu \propto$ to the momentum associated w/ the coordinates of M^D .

* Physical state conditions:

$$\psi[x^\mu(\sigma)] \equiv \langle x^\mu(\sigma) | \psi \rangle$$

which we express in terms of occupation #

basis via Creation ops

$$\psi[x^\mu(\sigma)] = \{ \phi(x^\mu) + iA_\mu^1(x^\mu) \alpha_1^\mu + iA_\mu^2(x^\mu) \alpha_2^\mu + \dots + \psi_0 \}$$

* Together form a basis in the Hilbert space

Differently. ψ_0 refers to the non-zero modes and satisfy

$$\alpha_n^M \psi_0 = 0, \quad n \geq 1$$

ψ_0 : vacuum state $\psi_0 \equiv \langle 0, 0 | \psi \rangle$

$$\therefore \alpha_n^M \psi_0 \equiv \langle 0, 0 | \alpha_n^M | \psi \rangle = 0$$

Since $\alpha_n^M |0, p\rangle = 0 \Leftrightarrow \langle 0, p | (\alpha_n^M)^\dagger = 0$
but $(\alpha_n^M)^\dagger = \alpha_{-n}^M$ so $n \rightarrow -n$ shows this.

* Claim: The vacuum eq for $|0, p\rangle$ is a position "basis" is

$$\langle X^M(\sigma) | 0, p \rangle = c \prod_{n=1}^{\infty} \exp \left(-\frac{n}{\alpha'} X_n^M X_{nM} \right)$$

* Recall that for the harmonic oscillator

$$-\frac{1}{2m} \frac{d^2}{dx^2} \psi + \frac{1}{2} m \omega^2 x^2 \psi = E \psi$$

$$\xi \equiv \sqrt{m\omega} x$$

$$\psi(\xi) = \left(\frac{m\omega}{2^n n! \pi} \right)^{1/4} H_n(\xi) e^{-\xi^2/2}$$

→ In this situation we have the Ground state

$\psi_0(x) \equiv \langle x | 0 \rangle$, which we find via

$$\langle x | \hat{a} | 0 \rangle = \langle x | \left(\sqrt{\frac{m\omega}{2}} \tilde{x} + i \sqrt{\frac{1}{2m\omega}} \tilde{p} \right) | 0 \rangle$$

but $\hat{a} | 0 \rangle = 0$

$$\langle x | 0 \rangle x \sqrt{\frac{m\omega}{2}} + i \sqrt{\frac{1}{2m\omega}} \left(-i \frac{d}{dx} \langle x | 0 \rangle \right) = 0$$

$$\psi_0 \times m\omega + \frac{d}{dx} \psi_0 = 0$$

$$\psi_0 = c \exp \left(-x^2 m\omega / 2 \right) \quad \begin{pmatrix} \text{Classical} \\ \text{Harmonic Oscillator} \end{pmatrix}$$

* For the open string: $\langle X^M(\sigma) | 0, p \rangle \equiv \psi_0$

$$\langle X^M | \alpha_n^M | 0, p \rangle = 0 \quad \hat{x}^M$$

$$2\alpha' \frac{d}{dx_{nn}} \psi_0 + \frac{n}{2} x_n^u \psi_0 = 0$$

$$\therefore \psi_{0,n} = c e^{-\left(\frac{n}{4\alpha'} x_n^u x_{nn}\right)}$$

This is for a single n , to obtain all the full state

$$\psi_0 = \prod_{n=1}^{\infty} \psi_{0,n} = c \prod_{n=1}^{\infty} \exp\left[-\frac{n}{4\alpha'} x_n^u x_{nn}\right]$$

(Vacuum state)

Physical states: $L_n \psi = 0$ ($\alpha_n^u = -i(\sqrt{2\alpha'} \frac{\partial}{\partial x_{nn}} + \frac{n}{\sqrt{2\alpha'}} \frac{x_n^u}{2})$)

$$\begin{aligned} L_0 &= \frac{1}{2} \alpha_0^u \alpha_{0,u} + \sum_{n=1}^{\infty} \alpha_n^u \alpha_{n,u} \\ &= \frac{1}{2} \left(-1 \left(\sqrt{2\alpha'} \frac{\partial}{\partial x_{0,u}} \right) \left(\sqrt{2\alpha'} \frac{\partial}{\partial x_{0,u}} \right) + \dots \right) \\ &= -\alpha' \left(\frac{\partial}{\partial x_0} \right)^2 - \sum_{n=1}^{\infty} \left(\sqrt{2\alpha'} \frac{\partial}{\partial x_{nn}} + \frac{n}{\sqrt{2\alpha'}} \frac{x_n^u}{2} \right) \left(\dots \right) \end{aligned}$$

$$L_0 \psi = L_0 e^{\phi + \dots} \psi_0$$

$$\begin{aligned} (\partial_1 + x_1)(\partial_2 + x_2)f &= (\partial_1 + x_1)(\partial_2 f + x_2 f) \\ &= \partial_1 \partial_2 f + \delta_{12} f + x_2 \partial_1 f + x_1 \partial_2 f + x_1 x_2 f \end{aligned}$$

In acting on ψ only the first two terms contribute (consider why.....)

$$L_0 e^{\phi + \dots} \psi_0 = -(\partial^2 + 1)\psi + \dots$$

Each variables term by term for us to find

$$(\alpha' \partial^2 + 1)\psi = 0 = (\alpha' \partial^2 - (5-1))A_n^S = (\partial^2 - 1)h_{\mu\nu} = \dots$$

as well as

$$\partial^\mu A_\mu^1 = 0 = -\sqrt{2\alpha'} \partial^\mu h_{\mu\nu} + A_\nu^2 = 2\sqrt{2\alpha'} \partial^\mu A_\mu^2 + h_\mu^4 = \dots$$

$$(\partial^2 - (D-2)/\alpha')^2$$

String Spectrum:

ϕ : Tachyon (negative mass) (Spin 0)
so E.M should be $(\partial^2 - m^2)$ for "normal" mass

A_μ^1 : Spin 1 - Vector

$h_{\mu\nu}$: Spin 2 Graviton

...

States at level n have a mass

$$m_n^2 = \left(\frac{2}{\alpha'}\right)(n-1)$$

- Particles of highest spin are contained in the contribution $\alpha_{-1}^{\mu_1} \dots \alpha_{-1}^{\mu_n} \epsilon_{\mu_1 \dots \mu_n}$ in ψ which have spin $J=n$

* We could impose $L_n \psi = 0$, $n \geq -1$, $\psi = e^{\phi} \psi_0$
obeys $\alpha' \partial^2 \phi = \dots$ i.e. where the "1's" are removed

$$L_{-1} \psi = 0 \Leftrightarrow \partial_\mu \psi = 0 = \partial_\mu A_\nu + \partial_\nu A_\mu$$

Soln: $\phi = (\text{constant})$

Remaining fields vanish

$$\begin{aligned} \psi[x^\mu(\sigma)] &= e^\phi \psi_0 = e^\phi \langle x^\mu(\sigma) | 0, p \rangle \\ &= \langle x^\mu(\sigma) | p \rangle \end{aligned}$$

Some insight into the Masses found:

* Classically $T_{\alpha\beta} = 0 \Leftrightarrow L_m = 0$, $m \in \mathbb{Z}$

$$L_0 = \tilde{L}_0 = 0.$$

$$\therefore (Mass)^2 =: M^2 = -p_\mu p^\mu$$

$$L_0 = \sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n + \alpha' p^2 = 0$$

$$M^2 = \frac{1}{\alpha'} \left(\sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n \right) \quad (\text{open string})$$

Physical states: $(L_0 - a)|\psi\rangle = 0$

$\leadsto$ Mass operator

$$\hat{N} \equiv \sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n$$

$$\alpha' M^2 = \hat{N} - a$$

Since we require $a=1$,

$$\alpha' M^2 = \hat{N} - 1$$

Ground state: $\hat{N} = 0$

$$\alpha' \hat{M}^2 = -1 \quad \text{Tachyon } \phi$$

$$\hat{N} = 1: \quad \alpha' \hat{M}^2 = 0$$

$$\hat{N} = 2: \quad \alpha' \hat{M}^2 = 1$$

* $X^\mu(\sigma, \tau)$ contains an ∞ # of possible neg.

all states, each associated to α_n^μ

$$[\alpha_n^\mu, \alpha_m^\nu] = -n \delta_{n,m} \delta^{\mu\nu}$$

ex: $\alpha_{-n}^\mu |0, p\rangle = (\alpha_n^\mu)^\dagger |0, p\rangle$

$$[\langle 0, p | \alpha_n^\mu \alpha_{-n}^\nu | 0, p \rangle = -n \langle 0, p | 0, p \rangle$$

- Physical states must have a positive norm if we are to interpret the formalism as a probability

(Prove later)

Thm: [No ghost Theorem] For space-time dim $D \leq 26$ states ψ which satisfy the eqs $L_n \psi = 0, n \geq 1, (L_0 - 1)\psi = 0$ have

* Bosare string inconsistent in $D \neq 26$

- For $D \neq 26$, the norms of the physical states are not all ≥ 0

ex: $|s\rangle \equiv L_{-1}|\Omega\rangle$ / $L_0|\Omega\rangle = 0$

$L_n|\Omega\rangle = 0, n \geq 1$

Claim: $L_0|s\rangle = |s\rangle$

Proof: $L_0 L_{-1}|\Omega\rangle = (L_{-1} + 0)|\Omega\rangle = |s\rangle \checkmark$

- Let $\langle\phi|$ be any other physical state, then we find

$\langle\phi|s\rangle = \langle\phi|L_{-1}|\Omega\rangle = (\langle\Omega|L_1|\phi\rangle)^*$

Since $L_1|\phi\rangle = 0$ (physical state requirement)

$\langle s|s\rangle = 0$

* Let $D = 26$:

$[L_n, L_m] = (n-m)L_{n+m} + \frac{D}{12}n(n^2-1)\delta_{n+m,0}$

$[L_n, L_m] = (n-m)L_{n+m} + \frac{13}{6}n(n^2-1)\delta_{n+m,0}$

• Take $|s'\rangle \equiv (L_{-2} + bL_{-1}^2)|\Omega'\rangle$

$L_n|\Omega'\rangle, n \geq 1$

$(L_0 + 1)|\Omega'\rangle = 0$

- Demanding $|s'\rangle$ be physical

$L_1|s'\rangle = 0$

$L_1(L_{-2} + bL_{-1}^2)|\Omega'\rangle$

$= (3L_{-1} + b2L_0L_{-1})|\Omega'\rangle$

$= (3L_{-1} + 2bL_{-2})|\Omega'\rangle$

or $(3 - 2b)L_{-2}|\Omega'\rangle$

$\Rightarrow b = \frac{3}{2}$

$n \geq 0$
 $L_n|\phi\rangle = 0$
 and
 $(L_0 - 1)|\phi\rangle = 0$
 Mass-shell condition

* zero-norm physical states $n \geq 1$

$$L_{-n_1} \dots L_{-n_p} |\Omega''\rangle,$$

$$L_n |\Omega''\rangle = 0$$

$$L_0 |\Omega''\rangle = \sum_{q=1}^p (n_q + 1) |\Omega''\rangle$$

SPECTRUM: Vector particle A^μ subject to Maxwell's Eq's:

$$\partial_\mu F^{\mu\nu} = \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$$

$$\partial^2 A^\nu - \partial_\mu \partial^\mu A^\nu = 0$$

• Choose the Gauge $\partial_\mu A^\mu = 0$

$$\underline{\partial^2 A^\nu = 0}$$

$$A^\mu(x^\sigma) = \int \frac{d^4k}{(2\pi)^4} (a^\mu_\lambda e^{-ikx} + a^{\mu\dagger}_\lambda e^{ikx})$$

$$a^\mu_\lambda \equiv \sum_{k=0}^3 a^\mu_\lambda e^k$$

λ : Polarization index:

$$A^\mu(x)|0\rangle \sim e^\mu_\lambda a^{\lambda\dagger}|0\rangle$$

which we find from the string spectrum if we identify

$$a^\mu_{-1} e^{ik_0 x^\sigma} \equiv a^{\mu\dagger}(k)$$

$$a^\mu_1 e^{-ik_0 x^\sigma} \equiv a^\mu(k)$$

$$A^\mu_m(x) = G_\mu e^{-ik_0 x^\sigma}$$

• $\exp(-ik_\mu x^\mu)$ is required to inject the momentum k^μ into the vacuum $|0\rangle$:

$$\exp(ik^\mu x_\mu)|0\rangle = |k\rangle$$

The E.O.M:

$$\partial^2 A^\mu = \partial^2 \int \frac{d^4k}{(2\pi)^4} (a^\mu_\lambda G_\lambda e^{-ikx} + a^{\mu\dagger}_\lambda e^{ikx}) = 0$$

$$\frac{d^4k}{(2\pi)^4} = k^3 dk \dots$$

and so we conclude

$$\boxed{k^2 = 0} \quad (\partial^2 A^\mu = 0)$$

* The Gauge choice tells us (trivially)

$$\boxed{K^\mu \epsilon_\mu = 0} \quad (\partial^\mu A_\mu = 0)$$

* In the Lorentz frame (Massless particle)

$$K^\mu = (k, 0, \dots, k)$$

$$K^\mu G_\mu = K G_0 + K G_{D-1} = 0$$

If $k \neq 0$, then $G_0 = -G_{D-1}$

$$\epsilon_\mu (\alpha_i^\mu)^\dagger |0, k\rangle = \begin{cases} (\alpha_i^\mu)^\dagger |0, k\rangle, & i=1, \dots, D-2 \\ (-\alpha_i^0)^\dagger + (\alpha_i^{D-1})^\dagger |0, k\rangle \\ (\alpha_i^0)^\dagger + (\alpha_i^{D-1})^\dagger |0, k\rangle \end{cases}$$

* Can remove zero norm state via the residual gauge symmetry

$$G_\mu \rightarrow G_\mu + K_\mu \lambda$$

- Leaves us with $(D-2)$ physical pos. def. norm states: matching our vector rep of $SO(D-2)$ (Little group)

* For $D=26$: the # of physical states w/ $k \cdot k > 0$ at level n , which have

$$\prod_{n=1}^{\infty} d(n) x^n = \prod_{n=1}^{\infty} \frac{1}{(1-x^n)^{24}} = 1 + 24x + \dots$$

$\underbrace{\quad}_{\phi} \quad \underbrace{\quad}_{A_\mu} \quad \dots$

ϕ : Has 9 D.O.F

A_μ : Has 24 D.O.F (in $D=26$)

$$\begin{aligned}
 p^\mu &= \frac{1}{2\pi\alpha'} = -i \frac{\partial x^\mu}{\partial \sigma} \\
 J^{\mu\nu} &= \frac{1}{2\pi\alpha'} \int_0^{2\pi} d\sigma (x^\mu \partial_\tau x^\nu - x^\nu \partial_\tau x^\mu) \\
 &= \frac{1}{2\pi\alpha'} \sqrt{\frac{\alpha'}{2}} \int_0^{2\pi} d\sigma (x^\mu \dot{x}^\nu - x^\nu \dot{x}^\mu) \\
 &= \frac{1}{\sqrt{2\alpha'}} (x_0^\mu \alpha_0^\nu - x_0^\nu \alpha_0^\mu) - i \sum_{n=1}^{\infty} (\alpha_{-n}^\mu \alpha_n^\nu - \alpha_n^\mu \alpha_{-n}^\nu) / n
 \end{aligned}$$

The Closed String:

We have $\dot{x}^\mu, p^\mu, \alpha_n^\mu, \bar{\alpha}_n^\mu$ with C.O.M coords (q^μ, p^μ) . Recall

$$\alpha_0^\mu = \bar{\alpha}_0^\mu = \sqrt{\frac{\alpha'}{2}} p^\mu$$

$$\{\alpha_n^\mu, \alpha_m^\nu\} = -in \delta_{n+m,0} \eta^{\mu\nu}, \quad \{\alpha_n^\mu, \bar{\alpha}_m^\nu\} = 0$$

$$\{\bar{\alpha}_n^\mu, \bar{\alpha}_m^\nu\} = -in \delta_{n+m,0} \eta^{\mu\nu}$$

→ Classical: From here we use Dirac Quant.

$$\begin{aligned}
 [\alpha_n^\mu, \alpha_m^\nu] &= n \delta_{n+m,0} \eta^{\mu\nu} & [\alpha_n^\mu, \bar{\alpha}_m^\nu] &= 0 \\
 [\bar{\alpha}_n^\mu, \bar{\alpha}_m^\nu] &= n \eta^{\mu\nu} \delta_{n+m,0} & [q^\mu, p^\nu] &= i \eta^{\mu\nu}
 \end{aligned}$$

For $n=0$:

$$\begin{aligned}
 \int_0^{2\pi} d\sigma \{p^\mu(\sigma), p^\nu(\sigma)\} &= \int_0^{2\pi} d\sigma \pi \eta^{\mu\nu} \delta(\sigma - \sigma') d\sigma' \\
 \int_0^{2\pi} d\sigma \{p^\mu(\sigma), \sqrt{2\alpha'} \pi p^\nu\} &= 2\pi \eta^{\mu\nu} \\
 \int_0^{2\pi} d\sigma \{p^\mu(\sigma), \sqrt{2\alpha'} \pi p^\nu\} &= (2\pi)^2 \eta^{\mu\nu} \\
 \text{use } \left\{ \frac{1}{2\alpha'} \left(\int_0^{2\pi} d\sigma \dot{x}^\mu + x^\mu \right), \sqrt{2\alpha'} \pi p^\nu \right\} &= (2\pi)^2 \eta^{\mu\nu} \\
 \left\{ \frac{1}{2\alpha'} (q^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau), \sqrt{2\alpha'} \pi p^\nu \right\} &= (2\pi)^2 \eta^{\mu\nu} \\
 \Rightarrow \{q^\mu, p^\nu\} &= \eta^{\mu\nu} \\
 \therefore [q^\mu, p^\nu] &= i \eta^{\mu\nu}
 \end{aligned}$$

* Reality conditions of $(\alpha_n^\mu)^\dagger = \bar{\alpha}_n^\mu$ implies:

$$(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu \quad | \quad \bar{\alpha}_{-n}^\mu = (\bar{\alpha}_n^\mu)^\dagger$$

$$q^{\mu\dagger} = q^\mu \text{ and } p^{\mu\dagger} = p^\mu$$

$$* \hat{P}^\mu = \frac{1}{2\pi\alpha'} \dot{x}^\mu, \quad \hat{P}^\mu = -i \frac{\delta}{\delta x_\mu(\sigma)}$$

$$P^H = \frac{1}{\sqrt{2\alpha'}} \left(2\alpha' P^H(0) + X'(0) \right) = \sum_{n=-\infty}^{\infty} \alpha_n^H e^{-in\sigma} \\ = \left(\sqrt{2\alpha'} \pi P^H + \frac{X'(0)}{\sqrt{2\alpha'}} \right)$$

$$\bar{P}^H = \left(\sqrt{2\alpha'} \pi \bar{P}^H - X'^H(0)/\sqrt{2\alpha'} \right) = \sum_{n=-\infty}^{\infty} \bar{\alpha}_n^H e^{in\sigma} \\ (3.1.6) \quad \& \quad (2.1.77)$$

$$X^H(\bar{c}, \sigma) = \sum_{n=-\infty}^{\infty} e^{in\sigma} \alpha_n^H(\tau) \\ \frac{\partial}{\partial X_n(0)} = \sum_{n=0}^{\infty} \frac{\partial X_n^H(0)}{\partial X_n(0)} \frac{\partial}{\partial X_n^H(0)} = \frac{1}{\pi} \sum_{n=0}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^H}$$

$$\sqrt{2\alpha'} \pi \left(\frac{1}{\pi} \sum_{n=0}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^H} \right) + \sum_{n=-\infty}^{\infty} -in e^{in\sigma} \alpha_n^H(\tau) / \sqrt{2\alpha'} \\ = \sqrt{2\alpha'} \frac{1}{2} \sum_{n=0}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^H} \\ \sum_{n=-\infty}^{\infty} e^{in\sigma} \left[\sqrt{\frac{\alpha'}{2}} \frac{\partial}{\partial X_n^H} + \frac{in \alpha_n^H}{\sqrt{2\alpha'}} \right] = \sum_{n=-\infty}^{\infty} \alpha_n^H e^{in\sigma}$$

$$\alpha_n^H = -i \sqrt{\frac{\alpha'}{2}} \frac{\partial}{\partial X_n^H} - \frac{in \alpha_n^H}{\sqrt{2\alpha'}}$$

$$\bar{\alpha}_n^H = -i \sqrt{\frac{\alpha'}{2}} \frac{\partial}{\partial X_{-n}^H} - \frac{in \alpha_n^H}{\sqrt{2\alpha'}}$$

Plugging these into $[\alpha_n^H, \alpha_m^H]$ we obtain the above.

* The Virasoro ops:

$$L_n = \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha_m^H \alpha_{n-m}^H : \mathcal{H}_{n,n}$$

$$\bar{L}_n = \frac{1}{2} \sum_{m=-\infty}^{\infty} \bar{\alpha}_m^H \bar{\alpha}_{n-m}^H : \mathcal{H}_{n,n}$$

Physical state conditions:

$$L_n \psi = \bar{L}_n \psi = 0, \quad n \geq 1$$

$$(L_0 + \bar{L}_0 - 2) \psi = 0$$

$$(L_0 - \bar{L}_0) \psi = 0$$

$\rightarrow$ Interpretation of $(L_0 - \bar{L}_0) \psi = 0$

$$L_0 = \frac{1}{2} : \sum_{n=-\infty}^{\infty} \alpha_n^\mu \alpha_{-n}^\mu : = \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + \sum_{n=1}^{\infty} \alpha_{-n}^\mu \alpha_n^\mu \eta_{\mu\nu}$$

$$(2.2.3) \text{ \& } (2.1.37)$$

$$x^\mu = \sum e^{in\sigma} \alpha_n^\mu(\tau)$$

$$(L_0 - \bar{L}_0) = \frac{2}{4\pi} \int_{-\pi}^{\pi} d\sigma ((\dot{X}^2) - (\dot{\bar{X}}^2)) = \frac{2}{4\pi} \int_{-\pi}^{\pi} d\sigma 2 \dot{X}^\mu(\sigma) \frac{\delta}{\delta X^\mu(\sigma)}$$

$2\pi\epsilon(L_0 - \bar{L}_0)$ implements the charge $\sigma \rightarrow \sigma + 2$ and so $(L_0 - \bar{L}_0)\psi = 0$ tells us there is no preferred pt. along the length of the closed string.

$$\psi[X^\mu(\sigma)] \equiv \langle X^\mu(\sigma) | \psi \rangle$$

$$\alpha_n^\mu \psi_0 = \bar{\alpha}_n^\mu \psi_0 = 0, \quad n \geq 1$$

$$(\alpha' \partial^2 + 4)\psi = 0$$

$$\partial^2 h_{\mu\nu} = 0$$

$$\partial^\mu h_{\mu\nu} = 0 = \partial^\nu h_{\mu\nu}$$

(c) Unwisted Strings: $\sigma \rightarrow -\sigma$

is a symmetry of the theory

• Every moving wave around the closed string which depends on $\tau + \sigma$ we have the same amplitude which depends on $\tau - \sigma$.

→ Equal amplitudes moving in opposite directions.

→ This imposes

$$\alpha_n^\mu \longleftrightarrow \bar{\alpha}_n^\mu$$

characterizing amplitudes for π^0 & η are unrelated.

BRST Symmetry